

The sidenotes package*

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Abstract

This package allows the typesetting of rich content in the margin. It includes text, but also figures, captions, tables and citations, which is common in science textbooks such as Feynman's *Lectures on Physics*.

Contents

1 Usage

- `\sidenote` The macro is very similar to the footnote macro and tries to emulate its behavior. It just puts the notes in the margin instead of the bottom of the page, therefore the name *sidenote*. It has the same parameters as footnote as well as an additional offset: `\sidenote[number][offset]{text}`. All the sidenotes are subsequently numbered and float in the margin to avoid overlap. The first, optional parameter will manually change the numbering to the given value. The second offset parameter will fix the text at a certain position in the margin. In particular, there is a difference between not providing an offset (floating text) and an offset of 0 (fix text at actual position).
- `\sidenotemark` Sidenote tries to mimic the footnote behavior and, consequently, provides the same solutions. Sometimes, it is not possible to directly call a sidenote macro, e.g. inside of a figure caption. Then, you can use `\sidenotemark[number]` and `\sidenotetext[number][offset]{text}` macros. `\sidenotemark` puts a mark at the current position. Afterwards, outside of the environment that causes the
- `\sidenotetext` trouble, it is possible the call `\sidenotetext[number][offset]{text}` to provide the text and typeset the sidenote. The optional parameters are similar to the sidenote macro. The first, optional parameter will change the numbering and the offset will change the position.
- `figure* (env.)` The `figure*` environment is used to position figures across the full page, i.e. the text width plus the margin. The algorithm has to distinguish between recto and verso (left and right) pages and might need up to three L^AT_EX runs to provide
- `table* (env.)` the desired result. The corresponding caption style is called *widefigure*. The sister environment for tables is `table*`. Use *widetable* to change its caption style.

*This document corresponds to sidenotes v1.20, dated 2024/09/12. Earlier versions by Oliver Schebaum.

2 Technical note

When writing the package, we tried to provide a *minimum* extension to standard L^AT_EX for typesetting rich content in the margin. This means, that there are no sensible default values for most things such as page geometry, fonts and font sizes. However, the *caesar_book*-class accompanies this package as an example implementation as well as a template we use for our theses.

In addition, we tried to keep compatibility with packages the user might want to use later. However, the following packages are needed by *sidenotes* and might introduce side effects with other packages.

3 Required packages

marginnote supports an alternative to `\marginpar` and creates notes in the margin. The notes are not floats and can be shifted up or down. Technically, every time an offset is provided the package uses `\marginnote` and `\marginpar` otherwise.

caption allows to set figure and table captions in the margin and allows easier formatting of these captions. Please refer to the *caption* manual for details on styles.

xparse is used to take advantage of the improved L^AT_EX3 syntax. All macros and environments are defined using this package.

l3keys2e provides a key/value mechanism

changepage is used to correctly shift figure* and table*. It has to use the option [strict] to work properly. This might lead to an option clash, if the same package is loaded without this option.

4 Implementation

Process the package options: onside

```
1 \ExplSyntaxOn
2 %
3 \keys_define:nn { sidenotes }
4 {
5   onside .bool_set:N = \sidenotes_onside
6 }
7 \ProcessKeysOptions { sidenotes }
```

We need a counter similar to the footnote counter.

```
8 \newcounter{sidenote} % make a counter
9 \setcounter{sidenote}{1} % init the counter
```

The L^AT_EX3 parts concerning the optional arguments should not distinguish between an empty bracket pair and no optional argument. This is required to keep macros with two leading optional arguments.

```
10 \DeclareExpandableDocumentCommand{\IfNoValueOrEmptyTF}{ m +m +m }
11 {
```

```

12 \IfNoValueTF{#1}
13   {#2}
14   {\tl_if_empty:nTF {#1} {#2} {#3}}
15 }

```

Put a marker in the horizontal list to detect subsequent sidenotemarks.

```

16 \NewDocumentCommand \@sidenotes@thesidenotemark { m }
17 {
18   \leavevmode
19   \ifhmode
20     \edef \@x@sf {\the \spacefactor }
21     \nobreak
22     \fi
23     \hbox {\@textsuperscript {\normalfont #1 }}
24     \ifhmode
25       \spacefactor \@x@sf
26     \fi
27     \relax
28 }
29 %
30 \NewDocumentCommand \@sidenotes@multisign { } {3sp}
31 %
32 \NewDocumentCommand \@sidenotes@multimarker { }
33 {
34   \kern-\@sidenotes@multisign
35   \kern\@sidenotes@multisign\relax
36 }
37 %
38 \NewDocumentCommand \@sidenotes@multichecker { }
39 {
40   \dim_compare:nNnTF \lastkern = \@sidenotes@multisign
41   {\@sidenotes@thesidenotemark{,}}
42   {}
43 }
44 %

```

Introduce an internal macro to place the marginal text. Use margin note in case an offset is given and marginpar otherwise.

```

45 \NewDocumentCommand \@sidenotes@placemarginal { +m +m }
46 {
47   \IfNoValueOrEmptyTF{#1}
48     {\marginpar{#2}}
49     {\marginnote{#2}[#1]}
50 }
51 %

```

\sidenote Introduce the `\sidenote` macro with two optional arguments to set the number and the offset.

```

52 \NewDocumentCommand \sidenote { o o +m }
53 {
54   \sidenotemark[#1]
55   \sidenotetext[#1][#2]{#3}
56   \@sidenotes@multimarker
57 }

```

`\sidenotemark` Sidenotemark is supposed to work similarly to footnotemark.

```

58 \NewDocumentCommand \sidenotemark { o }
59 {
60   \@sidenotes@multichecker
61   \IfNoValueOrEmptyTF{#1}
62     {\@sidenotes@thesidenotemark{\thesidenote}}
63     {\@sidenotes@thesidenotemark{#1}}
64   \@sidenotes@multimarker
65 }

```

`\sidenotetext` Sidenotetext is supposed to work similarly to footnotetext. The additional, optional argument sets the offset.

```

66 \NewDocumentCommand \sidenotetext { o o +m }
67 {
68   \IfNoValueOrEmptyTF{#1}
69     {
70       \@sidenotes@placemarginal{#2}{\textsuperscript{\thesidenote}{~#3}}
71       \refstepcounter{sidenote}
72     }
73     {\@sidenotes@placemarginal{#2}{\textsuperscript{#1}~#3}}
74 }

```

figure* (*env.*) The figure* environment provides a figure environment for figures that span across the full page (text plus margin width).

```

75 \AtBeginDocument{%
76 \newlength{\@sidenotes@extrawidth}
77 \setlength{\@sidenotes@extrawidth}{\marginparwidth}
78 \addtolength{\@sidenotes@extrawidth}{\marginparsep}
79 %
80 \NewDocumentEnvironment{autoadjustwidth}{ m m }{%
81 {
82   \bool_if:NTF \sidenotes_oneside
83   {
84     \begin{adjustwidth}{#1}{#2}
85   }
86   {
87     \begin{adjustwidth*}{#1}{#2}
88   }
89 }
90 {
91   \bool_if:NTF \sidenotes_oneside
92   {
93     \end{adjustwidth}
94   }
95   {
96     \end{adjustwidth*}
97   }
98 }
99 %
100 \DeclareCaptionStyle{widefigure}{font=footnotesize}
101 \RenewDocumentEnvironment{figure*}{ O{htbp} }
102 {
103   \begin{figure}[#1]

```

```

104         \begin{autoadjustwidth}{-}\@sidenotes@extrawidth}
105         \begin{minipage}{\linewidth}
106         \captionsetup{style=widefigure}
107     }
108 {
109 \end{minipage}
110     \end{autoadjustwidth}
111     \end{figure}
112 }

```

table* (*env.*) The table* environment provides a table environment for figures across text and margin width.

```

113 \DeclareCaptionStyle{widetable}{font=footnotesize}
114 \RenewDocumentEnvironment{table*}{ 0{htbp} }
115 {
116     \begin{table} [#1]
117         \begin{autoadjustwidth}{-}\@sidenotes@extrawidth}
118         \begin{minipage}{\linewidth}
119         \captionsetup{style=widetable}
120     }
121 {
122 \end{minipage}
123     \end{autoadjustwidth}
124     \end{table}
125 }
126 \ExplSyntaxOff

```